

KPI Guide: Formulas and Procedure

Support Agent Performance Analysis: how each dashboard KPI is calculated and built

Prepared for: Aaron | Data: tickets_raw_staging (1,924 tickets after cleaning) | Tools: MySQL 8.0 and Tableau

1. The Seven KPI Cards at a Glance

KPI card	Formula in plain words	Value
Total tickets	All tickets, summed over the 9 agents	1,924
Fixed by agent	Tickets the agent resolved or closed without escalating	1,280
Fixed rate	Fixed by agent divided by (fixed + escalated)	92.0%
Escalation rate	Escalated divided by (fixed + escalated)	8.0%
Avg. first response (hrs)	Average of first response time, weighted by each agent's tickets	12.1 hrs
Avg. resolution time (hrs)	Average resolution time, weighted by tickets that have one (fixed + escalated)	58.4 hrs
Avg. CSAT	Average CSAT, weighted by rated tickets (the fixed ones)	2.7 / 5

Supporting counts used inside the formulas but not shown on the cards: **still open 532** (Open and In Progress, not escalated) and **escalated 112**. Check: 1,280 + 532 + 112 = 1,924.

2. Where the Numbers Come From

Every KPI is built from one MySQL query, the agent scorecard, which returns one row per agent (9 rows).

Tableau reads it as a Custom SQL data source named **KPI_Tiles**. Each KPI then adds up or averages those nine rows.

Scorecard query for Tableau (no final ORDER BY, no comments, no semicolon)

```
WITH agent_fixed_escalation AS
(
  SELECT agent_id, assigned_agent,
    COUNT(CASE WHEN escalated = 'False'
      AND status IN ('Resolved', 'Closed') THEN 1 END) AS total_fixed,
    COUNT(CASE WHEN escalated = 'False'
      AND status IN ('Open', 'In Progress') THEN 1 END) AS total_still_open,
    COUNT(CASE WHEN escalated = 'True' THEN 1 END) AS total_escalated,
    COUNT(*) AS total_agent_handled,
    AVG(first_response_time_hrs) AS avg_frh,
    AVG(resolution_time_hrs) AS avg_rth,
    AVG(csatscore) AS avg_csatscore
  FROM tickets_raw_staging
  GROUP BY agent_id, assigned_agent
)
SELECT agent_id, assigned_agent, total_fixed, total_still_open, total_escalated,
  total_agent_handled,
  ROUND(avg_frh, 1) AS avg_response_time,
  ROUND(avg_rth, 1) AS avg_resolution_time,
  ROUND(avg_csatscore, 1) AS avg_csat_score
FROM agent_fixed_escalation
```

The fields the KPIs use are total_agent_handled, total_fixed, total_escalated, avg_response_time, avg_resolution_time and avg_csat_score. The status and escalated columns split each agent's tickets: a ticket is *escalated* when escalated is 'True' (its status is then Escalated), *fixed* when it is not escalated and its status is Resolved or Closed, and *still open* when its status is Open or In Progress.

3. One-Time Setup in Tableau

- **Connect:** Data menu, New Data Source, MySQL. Server localhost, port 3306, database customer_support, user root. If sign-in fails with "Authentication requires secure connection", log in once through MySQL Workbench first (or tick Require SSL and choose ca.pem as the SSL CA file).

- **Add the query:** double-click New Custom SQL, paste the scorecard query above, click Preview Results (expect **9 rows**) and OK. Rename the data source **KPI_Tiles**.
- **Create the five calculated fields** in Section 5 (right-click in the Data pane, Create Calculated Field).
- **Build one sheet** named KPIs (Section 6) and place it in the top row of the dashboard.

4. Each KPI: Meaning, Formula and Procedure

4.1 Total tickets

Item	Detail
What it tells you	How many tickets the team handled in the period.
Source columns	total_agent_handled (SQL: COUNT(*) per agent)
Tableau field	No calculated field. Drag total_agent_handled to the Measure Values card with the Sum aggregation.
Worked example	Priya S 247 + James T 215 + Lena K 194 + Omar F 216 + Sophie M 222 + Raj P 213 + Elena V 202 + Daniel N 212 + Yuki T 203 = 1,924.
Procedure	1. Add SUM(total_agent_handled) to the Measure Values card. 2. Right-click its column header, Edit Alias, type Total tickets. 3. Format: number, 0 decimals, thousands separator.
Check in SQL	SELECT COUNT(*) FROM tickets_raw_staging; (1,924)
Watch out	The cleaned table has 1,924 rows. The source dataset describes 2,000 tickets; 76 ticket IDs are missing from the original downloaded file.

4.2 Fixed by agent

Item	Detail
What it tells you	Tickets the agent resolved or closed without escalating them to engineering.
Source columns	total_fixed (SQL: COUNT(CASE WHEN escalated = 'False' AND status IN ('Resolved','Closed') THEN 1 END))
Tableau field	No calculated field. Drag total_fixed to Measure Values with the Sum aggregation.
Worked example	Priya S 163 + James T 139 + Lena K 136 + Omar F 149 + Sophie M 151 + Raj P 135 + Elena V 131 + Daniel N 141 + Yuki T 135 = 1,280.
Procedure	Add SUM(total_fixed) to Measure Values, then rename the header to Fixed by agent. Format: 0 decimals, thousands separator.
Check in SQL	SELECT COUNT(*) FROM tickets_raw_staging WHERE escalated = 'False' AND status IN ('Resolved','Closed'); (1,280)
Watch out	Not every non-escalated ticket is fixed. 532 are still open, which is why the first version of this count (escalated = 'False' alone) was overstated.

4.3 Fixed rate

Item	Detail
What it tells you	Of the tickets that reached an outcome, the share the agent fixed without escalating.
Tableau formula	SUM([total_fixed]) / (SUM([total_fixed]) + SUM([total_escalated]))
Why this denominator	Tickets that are still open have no outcome yet, so they are left out. The denominator is fixed + escalated = 1,392, so fixed rate + escalation rate = 100%.
Worked example	1,280 / (1,280 + 112) = 1,280 / 1,392 = 0.9195, shown as 92.0%.
Procedure	1. Create the calculated field Fixed rate. 2. Add it to Measure Values (it shows as AGG(Fixed rate)). 3. Right-click the field in the Data pane, Default Properties, Number Format, Percentage, 1 decimal.
Check in SQL	ROUND(100.0 * fixed / (fixed + escalated), 1) using the counts above (92.0)
Watch out	Do not type 100 * in the formula. The Percentage format already multiplies by 100, and doing both showed 804.6% on the escalation card.

4.4 Escalation rate

Item	Detail
What it tells you	Of the tickets that reached an outcome, the share handed to engineering. Used as a watch-list signal, so its number is shown in red.
Tableau formula	$\text{SUM}([\text{total_escalated}]) / (\text{SUM}([\text{total_fixed}]) + \text{SUM}([\text{total_escalated}]))$
Worked example	$112 / 1,392 = 0.0805$, shown as 8.0%.
Procedure	Same as Fixed rate: create the field, add it to Measure Values, format as Percentage with 1 decimal. For the red number, open the Format panel for AGG(Escalation rate), Pane tab, and set the font color to red.
Check in SQL	$\text{ROUND}(100.0 * \text{escalated} / (\text{fixed} + \text{escalated}), 1)$ (8.0)
Watch out	Across agents the rate runs from 4.5% to 11.2% on only 7 to 17 escalations each, so read it as a prompt to review tickets, not as a verdict.

4.5 Avg. first response (hrs)

Item	Detail
What it tells you	How long customers wait, on average, for the first reply.
Tableau formula	$\text{SUM}([\text{avg_response_time}] * [\text{total_agent_handled}]) / \text{SUM}([\text{total_agent_handled}])$
Why weighted	Each agent's average is based on a different number of tickets. Averaging nine averages would treat an agent with 203 tickets like one with 247. Multiplying each average by its ticket count and dividing by the total ticket count gives the true overall average.
Worked example	Weights are all of an agent's tickets, because every ticket has a first response time. Result 12.07, shown as 12.1 hrs.
Procedure	Create the field, add it to Measure Values, format: number, 1 decimal, suffix hrs (leading space).
Check in SQL	$\text{SELECT ROUND}(\text{AVG}(\text{first_response_time_hrs}), 1) \text{ FROM tickets_raw_staging;}$ (12.1)
Watch out	The SQL already rounds each agent average to 1 decimal, so the weighted result can differ from the exact value by about 0.02. It still displays as 12.1.

4.6 Avg. resolution time (hrs)

Item	Detail
What it tells you	How long it takes, on average, from creation until a ticket is resolved.
Tableau formula	$\text{SUM}([\text{avg_resolution_time}] * ([\text{total_fixed}] + [\text{total_escalated}])) / \text{SUM}([\text{total_fixed}] + [\text{total_escalated}])$
Why these weights	Resolution time exists only for fixed and escalated tickets (open tickets have none), so each agent's average is weighted by total_fixed + total_escalated, not by all tickets.
Worked example	Priya S: $163 + 13 = 176$ tickets with a resolution time, average 58.1 hrs. Yuki T: $135 + 14 = 149$ tickets, average 63.5 hrs. Priya's average counts for more. All nine agents give 58.4 hrs over 1,392 tickets.
Procedure	Create the field, add it to Measure Values, format: number, 1 decimal, suffix hrs.
Check in SQL	$\text{SELECT ROUND}(\text{AVG}(\text{resolution_time_hrs}), 1) \text{ FROM tickets_raw_staging;}$ (58.4). AVG skips the blank resolution times automatically.
Watch out	The simple average of the nine agent averages is 58.5 and is not the true value. The same mistake caused the wrong channel chart earlier (an average of category averages).

4.7 Avg. CSAT

Item	Detail
What it tells you	Average customer satisfaction on a 1 to 5 scale.
Tableau formula	$\text{SUM}([\text{avg_csat_score}] * [\text{total_fixed}]) / \text{SUM}([\text{total_fixed}])$
Why these weights	CSAT exists only for fixed tickets (Resolved and Closed). It is blank for every escalated ticket and every open ticket, so each agent's average is weighted by total_fixed.
Worked example	1,280 rated tickets, overall average 2.71, shown as 2.7 / 5.
Procedure	Create the field, add it to Measure Values, format: number, 1 decimal, suffix / 5 (with spaces, so it reads 2.7 / 5).
Check in SQL	<code>SELECT ROUND(AVG(csat_score), 1) FROM tickets_raw_staging; (2.7)</code>
Watch out	Escalations never lower CSAT, because escalated tickets have no rating. Mention this when presenting the number.

5. Calculated Field Reference

Field	Formula	Format
Fixed rate	$\text{SUM}([\text{total_fixed}]) / (\text{SUM}([\text{total_fixed}]) + \text{SUM}([\text{total_escalated}]))$	Percentage, 1 decimal
Escalation rate	$\text{SUM}([\text{total_escalated}]) / (\text{SUM}([\text{total_fixed}]) + \text{SUM}([\text{total_escalated}]))$	Percentage, 1 decimal
Avg first response (hrs)	$\text{SUM}([\text{avg_response_time}] * [\text{total_agent_handled}]) / \text{SUM}([\text{total_agent_handled}])$	Number, 1 decimal, suffix hrs
Avg resolution time (hrs)	$\text{SUM}([\text{avg_resolution_time}] * ([\text{total_fixed}] + [\text{total_escalated}])) / \text{SUM}([\text{total_fixed}] + [\text{total_escalated}])$	Number, 1 decimal, suffix hrs
Avg CSAT	$\text{SUM}([\text{avg_csat_score}] * [\text{total_fixed}]) / \text{SUM}([\text{total_fixed}])$	Number, 1 decimal, suffix / 5

The three weighted averages use the ticket counts as weights, so each agent's average counts in proportion to the tickets behind it.

6. Building the KPIs Sheet and Placing It on the Dashboard

- **New sheet** named KPIs, with the KPI_Tiles data source selected.
- **Columns:** drag Measure Names to Columns. **Text:** drag Measure Values to Text in the Marks card.
- **Measure Values card:** keep seven measures only: SUM(total_agent_handled), SUM(total_fixed), AGG(Fixed rate), AGG(Escalation rate), AGG(Avg first response), AGG(Avg resolution time) and AGG(Avg CSAT). Remove the rest (right-click, Remove).
- **Order:** drag the pills in the Measure Values card into the order of the cards.
- **Headers:** right-click each column header, Edit Alias, and enter the card names. Put units in the number suffix, not in the label, so "(hrs)" is not repeated.
- **Format:** right-click each pill in the Measure Values card, Format, Numbers, and apply the formats in Section 5.
- **Colors:** use one color for the card numbers. Select AGG(Escalation rate) in the Format panel, Pane tab, and set its font color to red.
- **Fit:** set the toolbar Fit dropdown to Entire View. Remove gridlines and borders (Format, Borders, None).
- **Dashboard:** in Tiled mode, drag the KPIs sheet into the top row. Turn its title off and give the row a fixed height of about 110 to 130 px.

7. Check All the KPIs With One Query

Run this in MySQL Workbench and compare it with the cards. It reads the raw table, not the scorecard, so it checks the whole chain. I tested the logic on the CSV; your MySQL output is the real confirmation.

```

SELECT COUNT(*) AS total_tickets,
SUM(escalated = 'False' AND status IN ('Resolved','Closed')) AS fixed_by_agent,
SUM(escalated = 'False' AND status IN ('Open','In Progress')) AS still_open,
SUM(escalated = 'True') AS escalated,
ROUND(100.0 * SUM(escalated = 'False' AND status IN ('Resolved','Closed'))
/ (SUM(escalated = 'False' AND status IN ('Resolved','Closed'))
+ SUM(escalated = 'True')), 1) AS fixed_rate_pct,
ROUND(100.0 * SUM(escalated = 'True')
/ (SUM(escalated = 'False' AND status IN ('Resolved','Closed'))
+ SUM(escalated = 'True')), 1) AS escalation_rate_pct,
ROUND(AVG(first_response_time_hrs), 1) AS avg_first_response_hrs,
ROUND(AVG(resolution_time_hrs), 1) AS avg_resolution_hrs,
ROUND(AVG(csat_score), 1) AS avg_csat
FROM tickets_raw_staging;

```

Column	Expected	Column	Expected
total_tickets	1,924	escalation_rate_pct	8.0
fixed_by_agent	1,280	avg_first_response_hrs	12.1
still_open	532	avg_resolution_hrs	58.4
escalated	112	avg_csat	2.7
fixed_rate_pct	92.0		

8. Troubleshooting

What you see	Cause	Fix
Escalation card shows 0	The value 0.0805 is shown as a whole number	Apply the Percentage format with 1 decimal to the field
Escalation card shows 804.6%	Multiplied by 100 twice (formula and format)	Remove 100 * from the formula, or use a plain number format
A response time of 108.6 hrs	Nine agents' averages were added together (SUM with no agent on Rows)	Use the weighted formula, or put the agent on Rows
Average is 58.5, not 58.4	Simple average of agent averages	Use the weighted formula with the ticket-count weights
Field shows a red ! (invalid)	The calculation refers to a column the data source does not have, often after copying a data source	Delete the copied field and recreate it in KPI_Tiles
Card is cut off on the right	The sheet is wider than its space	Set Fit to Entire View and give the row enough width
Tableau cannot sign in to MySQL	MySQL 8 login needs a secure first connection after a restart	Log in through Workbench first, or tick Require SSL with ca.pem

Values were verified against the raw CSV. All KPI formulas come from sessions with Claude and were checked on your own data.